

TinyOS For Your Low-Power IoT NEEDS

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The Internet of Things (IoT) phenomenon has truly caught the imagination of techies and developers worldwide. Companies, both non-profit and for-profit, are developing operating systems (OSes) for the same, promoting these as the best alternative to products already in the market.

In this article, we look at TinyOS, which just happens to be one of the most popular and oldest OSes specifically for low-power applications. TinyOS finds use in a multitude of applications, not restricted to wireless sensor networks, smartdust, personal area networks, automation in buildings, smartmeters and smart-whatever's connected to a local area network.

What began as a collaboration between UC Berkeley, Intel and Crossbow technology, has now become one of the most trusted and used OSes in this field and has even spawned an international consortium, the TinyOS Alliance. ESTCube-1, the world's first satellite with an electric solar wind sail, has its communication module powered by TinyOS. A humble beginning with a trip to space, what a story indeed!

In this month's DVD, you will find a copy of this software so you too can let your low-powered applications roar with life, may be even outside this planet.

What more is TinyOS

You may know that TinyOS is an open source, BSD-licensed OS

designed for low-power wireless devices, but what you may not know is that it provides software abstractions for underlying hardware. For example, when we might see a flash drive as a storage device, with certain read/write/erase properties, TinyOS looks at it as just another example of a circular log.

This allows it to simplify the job of application and system

it will be loaded on will have resource constraints in the form of a few kB of RAM and/or memory, or the need to run on low power.

The TinyOS-Arduino comparison

People sometimes wonder whether TinyOS and Arduino address the same need and, if so, if Arduino can replace TinyOS. The simple answer is 'No,' but let us not stop at giving simple responses.

What Arduino does is provide C support for MCUs and sensors, whereas TinyOS is more of a full-fledged OS with a host of developers constantly providing support and documentation. Ask any developer and they would swear that it is easier to get started from scratch in Arduino, and maybe build a simple application that runs

on a sensor. But where Arduino scores on ease and simplicity, in the long run, and when dealing with applications that require more processing power, TinyOS and its facilities in the area of networking, like multi-hop routing, reliable dissemination and time-synchronisation, do pay off huge.

There are also certain other factors to keep in mind, like the fact that while TinyOS is optimised

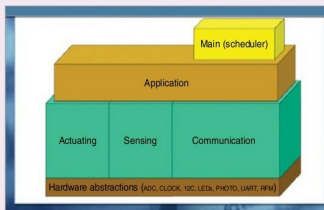


Fig. 1: Structure of TinyOS

developers as they can now use this sort of abstraction to reduce the time spent on one multi-step process by bringing it down to multiple single-step processes. We will discuss why this is critical in a bit.

TinyOS flourishes on microcontrollers (MCUs) with underlying sensors and/or network capabilities. It has been designed keeping in mind that most of the boards

What is RTOS and EOS

This discussion also brings out terms that are sometimes used interchangeably, not always retaining the meaning. An example that comes to mind is real-time operating system (RTOS) and embedded operating system (EOS). In reality though, while some EOSes do have real-time capabilities, not all can handle performing operations on real-time data.